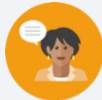


9.4 Writing a Quadratic Function From a Written Description

Lesson Introduction

 Benchmarks	MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.		 Learning Targets	- I can write a quadratic function to represent the relationship between two quantities from a written description. - I can rewrite a quadratic equation from vertex form to standard form. - I can solve for a variable of interest.	
 Benchmark Coherence	Building On N/A			Working Toward MA.912.AR.3.4 Write a quadratic function to represent the relationship between two quantities from a graph, a written description, or a table of values within a mathematical or real-world context; clarification 3.	
 Essential Questions	- How can I determine if a real-world scenario represents a quadratic function? - How can a quadratic function represent the relationship between two quantities?				
 Lesson Narrative	In this lesson, students will extend their knowledge of writing quadratic functions to real-world scenarios in the form of a written description. Students will examine real-world situations to translate the given information into the key information that's necessary for writing the quadratic function that represents the situation.				
 Component Summary	9.4.1 Warm-Up (~5 min.)	Graph and written description matching activity (<i>SE p. 77</i>)	9.4.3 Try It (10 min.)	Practice writing quadratic functions given written descriptions (<i>SE p. 79</i>)	
	9.4.2 Collaboration (15 min.)	Collaboration on writing quadratic functions given a written description (<i>SE p. 78</i>)	9.4.4 Your Turn! (10 min.)	Independent practice for writing quadratic functions given written descriptions (<i>SE p. 80</i>)	
	9.4.5 Wrap-Up (~5 min)	Sales pitch – which representation is easiest? (<i>SE p. 81</i>)			
 Resources	Glossary	Materials	Additional Preparation		
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> x-Intercept	N/A	N/A		

 Teacher Tips	Common Misconceptions	Things to Consider
	Students might misinterpret key features of quadratic functions when decoding word problems. Students may use the first number they see as a vertex, may always assume the lower value is always the x-value of the vertex, or other misconceptions born from gaps in decoding text.	Provide Three Reads routine scaffolding cards for students to leverage when decoding word problems. Providing scaffolding cards will give students additional entry points and lower the threshold of certain word problems so all students have a way to engage.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Think-Pair-Share 9.4.2 Collaboration provides students with a collaborative opportunity to work with other students to write quadratic functions given a written description, specific to an airline ticket scenario. Students will be able to think about the problems on their own, pair up with a partner, and share their thinking to learn from one another and come to a shared understanding.	MA.K12.MTR.5.1: Patterns and Structure Students will be working to write quadratic functions from a written description. To do this, students will need to focus on the relevant details within a problem.
	Lesson 4 provides a variety of opportunities to enhance real-world applications of quadratic functions. - Warm-Up: Prompt students to look ahead at this lesson’s Warm-Up after reviewing Lesson 3. Students can be in groups, tasked with the objective to bring in an object that illustrates that graph or find a video to display. - Wrap-Up: Sales pitch – challenge students to be creative in responses for the Wrap-Up. Instead of completing during class, put students in pairs and give them think time and planning time to design a creative sales pitch to be presented in the next lesson using technology, social media, or other resources.	
Lesson Notes:		

9.4.1 Warm-Up: Words, Tables, and Graphs - Part 2

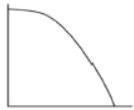
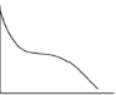
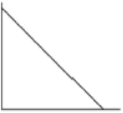
Component Overview: Students match graphs with the appropriate written description. This activity is an extension of the 9.3.1 Warm-Up from Lesson 3. This is designed to be done in pairs/groups. The conversation that results from this activity benefits student thinking.



9.4.1 Warm-Up: Words, Tables, and Graphs - Part 2

In the prior Warm-Up, the graphs were matched to a table of values.

- Match the same graphs to the appropriate written description.

Graphs	Written Description
I. 	A. The weight of your jumbo box of cereal decreases by an equal amount every week.
II. 	B. The tennis ball is dropped off the roof of a skyscraper.
III. 	C. For a while it looked like the decline in profits was slowing down, but then they began declining more rapidly.

I & B; II & C; III & A

- Discuss with your partner the strategy you used to match the verbal descriptions and graphs.
Answers will vary. I envisioned how quickly or slowly the situation declined and matched it to the appropriate graph. I looked for the linear situation and then the constantly rapid declining situation.



Have students describe what is happening in the graph and make connections between their responses here and their responses to the Warm-Up in the previous lesson. How are they the same? How are they different?

9.4.2 Collaboration: Airline Tickets

Component Overview: Students work together in pairs to collaborate to write quadratic functions given a written description.



9.4.2 Collaboration: Airline Tickets

An airline has a new plane with 120 seats and a new route from Tampa, FL to Cleveland, OH. The airline will maximize profit for this route at \$2,358 per flight. Based on a comprehensive customer survey, the airline determined that charging \$55 a seat will return a profit of \$2,100. They also found that charging \$75 a seat will return a profit of \$2,100. After analyzing all of the data, the airline determined that charging \$65 a seat would allow the company to maximize the profit.

1. Complete the statements.

The maximum profit for the route is ☐ \$2,100. ☒ \$2,358. The seat charge for customers when the airline has achieved the maximum profit is ☐ \$55. ☒ \$65. ☐ \$75.

The point at which the airline has maximized their profit is the vertex $(x, f(x))$. The x -value of the vertex is the seat charge at the point when the company has a maximum profit, where the maximum profit is the $f(x)$ value of the vertex.

2. The vertex in the context is ☐ (55, 2100) ☒ (65, 2358) ☐ (75, 2100) because this is the point where the ☐ customers will pay the most per seat. ☒ airline will make the most money.



Three Reads

Provide scaffolding for students to utilize the Three Reads routine in decoding word problems.

- *First Read:* What is happening? Read to understand the context.
- *Second Read:* What is the question asking you to do? Read to understand the purpose.
- *Third Read:* What information is important that might help answer the question? Read to find and mark key information.



Why might the airline not make as much money by charging more than \$65?



Why do you think it is helpful for a company to maximize their profit? What factors do you think would matter to a consumer (like you)?

9.4.2 Collaboration: Airline Tickets (continued)

The profit the airline makes is a function of the cost per seat. Because the profit has a maximum and two points that are equidistant from the vertex, a quadratic function can be used to model the context. In either standard form as $f(x) = ax^2 + bx + c$ or in vertex form $f(x) = a(x - h)^2 + k$, where the variable x is the cost per seat and f is the profit.

3. Discuss with your partner how to confirm that the two points (55,2100) and (75,2100) are equidistant from the vertex.

Answers will vary. The y-value of each point is the same. When graphed, the points are the same distance from the vertex in either direction.

4. Use the steps in the table to write the quadratic function that models the airline's profit.

Substitute values for the point $(x, f(x))$.	$f(x) = a(x - h)^2 + k$
Substitute values for the vertex (h, k) .	$f(x) = a(x - h)^2 + k$
Solve for the variable a .	$f(x) = a(x - h)^2 + k$

Write the equation for the quadratic function in vertex form.

$$\begin{aligned} 2,100 &= a(55 - h)^2 + k \\ 2,100 &= a(55 - 65)^2 + 2,358 \\ 2,100 &= a(-10)^2 + 2,358 \\ 2,100 - 2,358 &= a(100) \\ -258 &= a(100) \\ a &= -2.58 \end{aligned}$$

$$f(x) = -2.58(x - 65)^2 + 2,358$$



Question 3 is more conceptual in nature, while question 4 is more procedural. Use student performance on these two questions to better understand what additional support students may need to master this standard or what small groups would look like during Try It or Your Turn!

Question 3: Students may not understand the word "equidistant" and need additional support on what that means and what it means in context. For ELL students, there may be a language barrier.

Question 4: The following misconceptions may exist:

- Incorrect input values into the function
- Miscalculation in determining the value of a
- Incorrect writing of a function with the correct value of a



Watch for calculation errors in question 4. If a student gets the wrong answer, check to see if their input is correct. If it is, check their calculations for the value of a .

9.4.3 Try It: Writing Quadratic Functions

Component Overview: Students will work to write quadratic functions given written descriptions. This activity is designed to be done with a partner pairing. Consider pairings based on ability or learning style. This activity is word-problem based, so consider reading abilities when developing pairs.

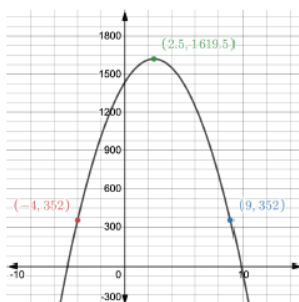


9.4.3 Try It: Writing Quadratic Functions

Work with your partner on the following.

- Frieda is creating a quadratic equation to represent the number of movie tickets sold in the U.S., in millions, from 2000 to 2005. She determines the number of tickets sold, T , is a function of the year.

x	$T(x)$
-4	352
-3.5	539.5
2.5	1619.5
4.5	1499.5
9	352



Write the quadratic function in both vertex and standard form that models the context. Show your work. Then, write the equations in the table in problem 3.

$$y = a(x - 2.5)^2 + 1,619.5$$

$$352 = a(9 - 2.5)^2 + 1,619.5$$

$$352 = a(6.5)^2 + 1,619.5$$

$$-1,267.5 = a(42.25)$$

$$a = -30$$

$$y = -30(x - 2.5)^2 + 1,619.5$$

$$y = -30x^2 + 150x + 1432$$



How can you interpret the table and graph based on the context? What does each point from the table mean in terms of the context?



For students who finish early, consider having them determine another point on the parabola that is not listed in the table. Have them prove their point algebraically.

9.4.3 Try It: Writing Quadratic Functions (continued)

2. A botanist was monitoring the growth of a type of tree during its first year. The botanist measured the heights of the trees in micrometers and the amount of rain they received in inches.

The botanist discovered the trees grew to an average height of 90,671 micrometers when there was 1 inch of rain or 43 inches of rain. The trees averaged the tallest height of 245,462 micrometers when there was 22 inches of rain.

Write the equation for the quadratic function of the average height of this type of tree in micrometers, H , as a function of the inches of rain, x , in vertex and standard form. Show your work below. Then, write the equations in the table in problem 3.

$$y = a(x - 22)^2 + 245,462$$

$$90,671 = a(1 - 22)^2 + 245,462$$

$$90,671 = a(-21)^2 + 245,462$$

$$-154,791 = a(441)$$

$$a = -351$$

$$y = -351(x - 22)^2 + 245,462$$

$$y = -351x^2 + 15,444x + 75,578$$

3. Ask another pair of students to check your work for the two quadratic equations you wrote for each context, while you and your partner check their work. If necessary, correct any errors. Each person should write their initials next to your equations stating that your equations are correct or were corrected after fixing any errors.

Context	Vertex Form	Standard Form	Initials
Movie Tickets	$y = -30(x - 2.5)^2 + 1,619.5$	$y = -30x^2 + 150x + 1432$	
Tree Height	$y = -351(x - 22)^2 + 245,462$	$y = -351x^2 + 15,444x + 75,578$	



The values in question 2 are larger than what they have worked with in this lesson. Remind students to pace themselves during the calculations and use the appropriate tools, such as a calculator, to check their work.



Consider having a “mistakes showcase” activity at the end of question 3. Have students share out mistakes that they made in the process that became clear after checking with another partner pair. Use growth-mindset language to encourage the building of knowledge and learning from mistakes.

9.4.4 Your Turn!

Component Overview: Students will work to write quadratic functions given written descriptions. This activity is designed to be done independently to allow for a practice opportunity for students.



9.4.4 Your Turn!

- Izaak and Mylee just bought a house with open land in the backyard. The previous owner left 400 linear feet of fencing. Izaak and Mylee want to construct the fence to give their two dogs the most area possible to play in the backyard. If they make the length of the fence 80 linear feet or 120 linear feet, the dogs will have a play area of 9,600 square feet. However, if Izaak and Mylee make the length of the fence 100 linear feet, they will be able to create a maximum area of 10,000 square feet.

The area of play for the dogs, A , in square feet is a function of the length of the fence, x , in linear feet.

- What is the point (x, y) where the dogs will have the maximum fenced in area?

(100, 10000)

- Identify one other point from the context.

Vertex (h, k)	Point $(x, A(x))$
(100, 10000)	(80, 9600)

- Using the vertex and point, write the quadratic function in vertex form.

Vertex Form	$y = a(x - 100)^2 + 10000$ $9600 = a(80 - 100)^2 + 10000$ $9600 = a(-20)^2 + 10000$ $-400 = a(400)$ $a = -1$ $y = -1(x - 100)^2 + 10000$
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Three Reads

Provide scaffolding for students to utilize the Three Reads routine in decoding word problems.

- First Read:* What is happening? Read to understand the context.
- Second Read:* What is the question asking you to do? Read to understand the purpose.
- Third Read:* What information is important that might help answer the question? Read to find and mark key information.



How can you determine another point from the context?
How did you determine which value would be the x -coordinate and which value would be the y -coordinate?



Challenge students to write the equation in standard form.

9.4.4 Your Turn! (continued)

2. An Olympic diver is preparing for his upcoming diving competition. He must execute his dive off of a 10 foot high diving board. In order to have enough time to execute all the twists and flips, after 2 seconds he needs to be at a height of 14 feet. After 4 seconds, he needs to be back at 10 feet above the water for his last flip as he finishes his dive.

The diver's height in feet, H , is a function of time in seconds, x .

- a. Identify the vertex and one additional point of the function from the context.

Vertex (h, k)	Point $(x, H(x))$
$(2, 14)$	$(4, 10)$

- b. Write the quadratic function in standard form.

$$\begin{aligned}
 y &= a(x - 2)^2 + 14 \\
 10 &= a(4 - 2)^2 + 14 \\
 10 - 14 &= a(2)^2 \\
 -4 &= a(4) \\
 a &= -1
 \end{aligned}$$

$$\begin{aligned}
 y &= -1(x - 2)^2 + 14 \\
 y &= -x^2 + 4x - 4 + 14 \\
 y &= -x^2 + 4x + 10
 \end{aligned}$$



This question asks students to write in standard form. There is no reference to vertex form first. Students may have misconceptions of how to use the value of a and go straight to standard form.

9.4.5 Wrap-Up: Cool Down

Component Overview: Students will complete a sales pitch to convince a classmate which representation is easiest for writing quadratic functions. This activity is designed to be done independently to demonstrate conceptual understanding from the first four lessons on the unit.



9.4.5 Wrap-Up: Cool Down

The last three lessons have involved writing quadratic functions to represent the relationship between two quantities from a graph, a table, and a written description.

1. Create a sales pitch to convince your classmates which representation you feel is the easiest for writing quadratic functions.

Answers will vary.



This open-ended activity can provide an insight into what students are able to articulate about the first three lessons of the unit. Consider this a summative response for those three lessons to inform what differentiation might look like for Lesson 5.